

Context Compliance System (CCS)

Comprehensive Rule Definitions, Parameters, and System Architecture

1. Core Context Compliance Rules

Rule ID	Rule Name	Condition Checked	Violation Trigger
CCR-1	Context Matching Rule	Observed data flow must match declared policy	Any mismatch
CCR-2	Undeclared Recipient Rule	All contacted domains must be declared	Hidden third-party detected
CCR-3	Consent Enforcement Rule	If consent denied → no storage/transmission	Data sent after denial
CCR-4	Derived Data Declaration Rule	Derived attributes must be declared	Derived data undisclosed
CCR-5	Data Minimization Rule	Data must match purpose	Excessive data collection

Rule No.	Rule Name	Condition (IF)	Action (THEN)	Dataset Used	Purpose
1	Purpose Matching Rule	Actual usage purpose ≠ Consented purpose	Flag Violation	context_compliance_dataset.csv	Train/Test compliance model
2	Permission Scope Rule	Accessed permission ≠ Granted permissions	Flag Violation	googleplay-app-permission.json	Permission comparison logic
3	Over-Collection Rule	Data fields collected > Required for stated purpose	Flag Warning/Violation	context_compliance_dataset.csv	Define minimal data guidelines
4	Time Validity Rule	Data used after consent expiry	Flag Violation	context_compliance_dataset.csv	Temporal compliance checking
5	Third-Party Sharing Rule	Data shared with unapproved entity	Flag Violation	context_compliance_dataset.csv	Sharing compliance logic
6	Frequency Abuse Rule	Access count > Normal threshold	Flag Suspicious	breaches.csv	Risk threshold modeling
7	Sensitive Data Rule	Sensitive data used without explicit consent	Critical Violation	context_compliance_dataset.csv	Severity classification
8	Context Drift Rule	Usage purpose gradually deviates over time	Flag Drift Alert	context_compliance_dataset.csv	Advanced monitoring

🤖 Rule Type Classification (Rule-based vs ML-based)

Rule No.	Rule Name	Best Approach	Why
1	Purpose Matching Rule	ML + NLP	Consent text aur usage text compare karna hai → similarity model better
2	Permission Scope Rule	Rule-Based	Simple set comparison (granted vs accessed)
3	Over-Collection Rule	Rule-Based	Predefined minimal data mapping se compare
4	Time Validity Rule	Rule-Based	Date comparison logic
5	Third-Party Sharing Rule	Rule-Based	Approved entity list se matching
6	Frequency Abuse Rule	ML (Anomaly Detection)	Normal vs abnormal access pattern detect karna
7	Sensitive Data Rule	Rule-Based + Severity Tag	Sensitive fields predefined hote hain
8	Context Drift Rule	ML (Semantic Drift / Similarity Over Time)	Purpose change gradual hota hai → NLP needed

2. Data Classification Parameters

Parameter	Description	Example	Usage in CCS
data_type	Category of collected data	IP, cookie	Identify personal data
sensitivity_level	Risk level	Low/Medium/High	Risk scoring
identifiability_score	Probability of identifying user	0–1 scale	GDPR applicability
linkability_score	Can combine with other data	High if fingerprint	Profiling detection
derived_flag	Whether data is inferred	City from IP	Check disclosure compliance

3. Consent Monitoring Parameters

Parameter	Description	Possible Values	Compliance Use
consent_given	User accepted?	True/False	Core decision variable
consent_scope	Consent coverage	Ads/Analytics/ML	Match with purpose
consent_timestamp	Time of consent	Datetime	Validity check
consent_source	Where consent collected	Banner/CMP	Trace accountability
consent_state_at_request	Consent at time of request	Granted/Denied	Detect violations

4. Data Flow Verification Structure

Observed Component	Declared Component	CCS Action
Observed Recipient Domain	Policy-listed domain	Compare
Observed Data Type	Declared Data Type	Match
Observed Purpose	Declared Purpose	Validate
Observed Cookie	Declared Cookie	Verify
Observed Processing	Declared Processing	Flag mismatch

5. Provenance Tracking Structure

Element	Definition in CCS	Example
Entity	Data item	User_IP
Activity	Processing step	Feature Extraction
Agent	Responsible party	Website / ML Model
used	Activity used entity	Training used User_Data
wasGeneratedBy	Data created by activity	Model_Output generated by Training
wasAttributedTo	Data linked to agent	Gradient linked to Client

6. ML Lifecycle Compliance Rules

Stage	CCS Checks	Violation If
Data Preprocessing	Secure extraction?	Raw data exposed
Model Training	Encrypted gradients?	Plaintext gradients
Poisoning Defense	Malicious filtering?	No anomaly detection
Model Inference	Encrypted inference?	Input leakage
Nonlinear Evaluation	Secure approximation?	Undisclosed accuracy loss

7. Third-Party Risk Detection

Risk Type	Detection Logic	Severity
Hidden Third Party	Domain not declared	High
Third-Party Cookie	Undeclared cookie set	High
Referral URL Leak	URL shared externally	Medium
Passive Data Leak	IP + UA silently sent	Medium
Consent Bypass	Data sent despite denial	Critical

8. Risk Scoring Model

Risk Factor	Weight
Undeclared Sharing	5
Consent Violation	5
Personal Data Exposure	4
Untraceable Provenance	3
Excessive Data Collection	3
Derived Undeclared Data	4